Algebraic Topology

Notes made by Finley Cooper

30th October 2026

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1 Introduction

1.1 Recap of IB Topological Spaces

We use algebra to turn topological questions to algebraic questions. This results in a loss of information but lets us prove some results.

Example.

$$\mathbb{R}^n \cong \mathbb{R}^m \implies n = m$$

This is a statement which is very hard to prove otherwise. Likewise we can distinguish topological surfaces with a different number of 'holes' in their surface.

For this section all functions are continuous until mentioned otherwise.

Definition. (Map) A *map* is a continuous function.

Lemma. (Gluing lemma) Let $f : X \to Y$ be a function and $C, K \subset X$ be closed subsets with $C \cup K = X$. Then f is continuous if and only if the restriction of f on C and K is continuous.

Proof. We have that f is continuous if and only if for all closed sets $D \subset Y$ the preimage of D is closed. Since $f^{-1}(D) = (f^{-1}(D) \cap C) \cup (f^{-1}(D) \cap K)$, this holds if and only if $\forall D \subset Y$ closed we have that $f^{-1}(D) \cap C$ and $f^{-1}(D) \cap K$ are closed. Since C and K are closed in X, a subset of C (resp. K) is closed in C (resp. K) if and only if it is closed in X. Hence this holds if and only if $(f|_C)^{-1}(D), (f|_K)^{-1}(D)$ are closed in C and K respectively. Hence this holds if and only if the restriction of f on C and K is continuous. □

Lemma. (Lebesgue number lemma) Let (X, d) be a compact metric space. For any open cover $\mathcal{U} = \{U_\alpha\}_{\alpha \in I}$ of X there exists a $\delta > 0$ such that for all $x \in X$, $B_\delta(x) \subset U_\alpha$ for some $\alpha \in I$.

Proof. Suppose no such δ existed. Then for all $n > 0$ there is some $x_n \in X$ such that $B_{1/n}(x_n) \not\subset U_\alpha$ for all $\alpha \in I$. The sequence $\{x_n\}$ admits a convergent subsequence due to compactness, so *wlog* replace x_n with its convergent subsequence, and take $x_n \to y$. Then take $\alpha \in I$ such that $y \in U_\alpha$. Then since U_α is open there exists a $r > 0$ such that $B_r(y) \subset U_\alpha$. Take $N > 0$ such that

$$\begin{cases} \frac{1}{N} < \frac{r}{2} \\ d(x_N, y) < \frac{r}{2} \end{cases}$$

which implies that $B_{1/N}(x_N) \subset B_r(y) \subset U_\alpha$ which is a contradiction. □

1.2 Cell complexes

Definition. (Cell attachment) Given a space X and a map $f : S^{n-1} \to X$. The space $X \cup_f D^n := X \sqcup D^n / \sim$ where $x \sim f(x), \forall x \in S^{n-1} \subset D^n$ is the result of attaching an n-cell to X along f.
The image of f is called the *attaching sphere*.

Definition. (Cell complex) A *finite-dimensional cell complex* is obtained by:

(i) Start with a set X^0 with the discrete topology.

(ii) Inductively once X^{n-1} is defined we form X^n by attaching a collection of n-cells along some collection of attachment maps

$$\{f_\alpha : S^{n-1} \to X^{n-1}\}_{\alpha \in I}.$$

i.e. that $X^n = \left(X^{n-1} \sqcup \bigsqcup_{\alpha \in I} D^n_\alpha\right) / \sim$ where we define $x \sim f_\alpha(x)$ for $x \in S^{n-1}_\alpha = \partial D^n_\alpha$.

(iii) $X = X^k$ for some $k \geq 0$. The minimal such k is called the dimension of X.

Remark. In general we can make sense of infinite dimensional cell complexes. Suppose that we have

$$X^0 \subset X^1 \subset \cdots$$

such that

(i) X^0 is discrete,

(ii) X^n is formed by attaching n-cells to X^{n-1}.

Define

$$X = \bigcup_{n \geq 0} X^n$$

with the topology where $U \subset X$ is open $\iff U \cap X^n$ is open in X^n for all n. We say that X is a CW-complex and that X^n is the n-skeleton of X.

2 Homotopy and the fundamental group

2.1 Homotopy

Definition. We denote by I the closed interval $[0, 1]$.

Definition. (Homotopy) Let $f, g : X \to Y$. Then we say that a map $H : X \times I \to Y$ is a *homotopy from f to g* if $H(x, 0) = f(x)$ and $H(x, 1) = g(x)$. If such an H exists we say that f is *homotopic* to g and $f \simeq g$.

If $A \subset X$ and $f|_A = g|_A$ then a homotopy H from f to g is relative to A (i.e. a homotopy rel A) if $H(a, t) = f(a) = g(a)$ for all $a \in A$ and $t \in I$. If such an H exists we write $f \simeq_A g$.

Proposition. The relation homotopic rel A defines an equivalence relation.

Proof.

(i) $f \simeq_A f$ trivially by $H(x, t) = f(x)$.

(ii) $f \simeq_A g \implies g \simeq_A f$, by replacing t with $1 - t$.

(iii) Suppose that $f \simeq_A g, g \simeq_A h$. Then let H_1 be the homotopy rel A from f to g and H_2 for g to h. Then define

$$H(x,t) = \begin{cases} H_1(x,2t) & 0 \le t \le \frac{1}{2} \\ H_2(x,2t-1) & \frac{1}{2} \le t \le 1 \end{cases}.$$

Since $H_1(x,1) = g(x) = H_2(x,0)$, H is well-defined, and by the gluing lemma (applied to the closed sets $X \times [0, \frac{1}{2}]$ and $X \times [\frac{1}{2}, 1]$) H is continuous. Also $H(a,t) = f(a)$ for all $a \in A$, so H is a homotopy rel A from f to h and therefore $f \simeq_A h$. □

Definition. (Homotopy equivalence) Given a map $f : X \to Y$ we say that f is a *homotopy equivalence* if there exists a $g : Y \to X$ such that $f \circ g \simeq \mathrm{id}_Y$ and $g \circ f \simeq \mathrm{id}_X$. We call g a *homotopy inverse* of f. If there exists a homotopy equivalence $X \to Y$ then X is *homotopy equivalent* to Y and we write that $X \simeq Y$.

Definition. (Contractible) A space X is *contractible* if X is homotopy equivalent to the one-point space $\{*\}$.

Lemma. Let $f_0, f_1 : X \to Y$ be homotopic maps and $g_0, g_1 : Y \to Z$ also be homotopic maps. Then

$$g_0 \circ f_0 \simeq g_1 \circ f_1.$$

Proof. We want to show that $g_0 \circ f_0 \simeq g_0 \circ f_1 \simeq g_1 \circ f_1$.

Choose homotopies H from f_0 to f_1 and H' from g_0 to g_1. Then $g_0 \circ H$ is a homotopy from $g_0 f_0$ to $g_0 f_1$. We also have that $H' \circ (f_1 \times \mathrm{id}_I)$ is a homotopy from $g_0 f_1$ to $g_1 f_1$ hence we're done by transitivity. □

Proposition. Homotopy equivalence is an equivalence relation.

Proof.

(i) $X \simeq X$ is obvious from the identity map.

(ii) $X \simeq Y$ then we have $Y \simeq X$ just by switching the roles of f and g.

(iii) $X \simeq Y$ and $Y \simeq Z$. Then suppose f, g are the homotopy equivalence and its inverse for X and Y, and f', g' for Y and Z. Consider the map $f'f$ from X to Z and its candidate homotopy inverse gg'. By the lemma above,

$$gg'f'f \simeq g\mathrm{id}_Y f = gf \simeq \mathrm{id}_X$$
$$f'fgg' \simeq f'\mathrm{id}_Y g' = f'g' \simeq \mathrm{id}_Z$$

Therefore $f'f$ is a homotopy equivalence and $X \simeq Z$. □

Example. Consider the inclusion

$$\tau : S^1 \hookrightarrow \mathbb{R}^2 \setminus \{0\}$$

with homotopy inverse $r : \mathbb{R}^2 \setminus \{0\} \to S^1$ given by projecting each vector onto its unit vector, $r(x) = x/\|x\|$. Since vector normalisation doesn't do anything to unit vectors one of the directions of composition is obvious: $r \circ \tau = \mathrm{id}_{S^1}$. For the other direction,

$$\mathbb{R}^2 \setminus \{0\} \to S^1 \hookrightarrow \mathbb{R}^2 \setminus \{0\}$$

we're given the homotopy

$$\begin{aligned} H : (\mathbb{R}^2 \setminus \{0\}) \times I &\to \mathbb{R}^2 \setminus \{0\} \\ (x, t) &\mapsto \frac{x}{t + (1-t)\|x\|} \end{aligned}$$

which is well-defined since the denominator is strictly positive, and is a homotopy from $\tau \circ r$ to $\mathrm{id}_{\mathbb{R}^2 \setminus \{0\}}$. Therefore we have that $S^1 \simeq \mathbb{R}^2 \setminus \{0\}$. Similarly we have that $S^{n-1} \simeq \mathbb{R}^n \setminus \{0\}$.

Example. We have that $\mathbb{R}^n$ is contractible. Let $f : \mathbb{R}^n \to \{0\}$ be the constant map and $g : \{0\} \hookrightarrow \mathbb{R}^n$ be the inclusion. Then

$$f \circ g = \mathrm{id}_{\{0\}}$$

so one direction is trivial. For the other direction, $g \circ f : \mathbb{R}^n \to \mathbb{R}^n$ is the constant map $x \mapsto 0$. Define the straight-line homotopy

$$\begin{aligned} H : \mathbb{R}^n \times I &\to \mathbb{R}^n \\ (x, t) &\mapsto tx. \end{aligned}$$

This is continuous since scalar multiplication $\mathbb{R} \times \mathbb{R}^n \to \mathbb{R}^n$ is continuous. We have $H(x, 0) = 0 = (g \circ f)(x)$ and $H(x, 1) = x = \mathrm{id}_{\mathbb{R}^n}(x)$, so H is a homotopy from $g \circ f$ to $\mathrm{id}_{\mathbb{R}^n}$. Therefore $g \circ f \simeq \mathrm{id}_{\mathbb{R}^n}$, so f is a homotopy equivalence with homotopy inverse g. Hence $\mathbb{R}^n \simeq \{0\}$, i.e. $\mathbb{R}^n$ is contractible.

The same argument works for any $X \subset \mathbb{R}^n$ which is star-shaped about some point x_0 (e.g. any convex set), using $H(x, t) = (1 - t)x_0 + tx$.

Definition. (Deformation Retraction) Let $i : A \hookrightarrow X$ be the inclusion of a subspace. We say that A is a *deformation retract* of X if there is a map $r : X \to A$ such that $r \mid_A = \mathrm{id}_A$ and $i \circ r \simeq \mathrm{id}_X$. If the homotopy is relative to A, then A is a strong deformation retract of X.

Remark. This implies that $X \simeq A$ by the inclusion map from A to X and r from X to A. This is a strictly stronger condition.

Definition. (Retraction) A *retraction* of X to A is a map $r : X \to A$ such that $r \mid_A = \mathrm{id}$.

Remark. A retraction need not be a homotopy equivalence.

Example. X is a space and $x \in X$. Then X retracts to $\{x\}$. But $X \simeq \{x\} \iff X$ is contractible

2.2 Paths

Definition. (Path) For a space X and points $x_0, x_1 \in X$, a *path* from x_0 to x_1 is a map $\gamma : I \to X$ such that $\gamma(0) = x_0$ and $\gamma(1) = x_1$.

We write that $\gamma : x_0 \rightsquigarrow x_1$, and if $x_0 = x_1$ we say that γ is a loop based at x_0.

If we have $\gamma : x_0 \rightsquigarrow x_1$ and $\gamma' : x_1 \rightsquigarrow x_2$ then the concatenation of $\gamma \circ \gamma'$ is defined by using the gluing lemma and going in double time which goes from x_0 to x_2.

We also define the inverse path of $\gamma : x_0 \rightsquigarrow x_1$ as $\gamma^{-1}(t) = \gamma(1-t) : x_1 \rightsquigarrow x_0$.

Finally we define the constant path $c_{x_0}(t) = x_0 : x_0 \rightsquigarrow x_0$.

If we define a relation $\sim$ on X by $x \sim y \iff$ there is a path from x to y, these definitions show that $\sim$ is an equivalence relation.

Definition. (Path components) The equivalence classes of $\sim$ are called *path components* of X. The set of all path components of X is denoted by $\pi_0(X)$. If $|\pi_0(X)| = 1$ then we say that X is *path-connected*.

Remark. X is path-connected $\iff$ $\forall x, y \in X$ there exists a path $x \rightsquigarrow y$.

Proposition. Given a map $f : X \to Y$ there is a well-defined associated function

$$\pi_0(f) : \pi_0(X) \to \pi_0(Y)$$

given by $\pi_0(f)[x] = [f(x)]$. Futhermore,

(i) If $f \simeq g$, then $\pi_0(f) = \pi_0(g)$;

(ii) For maps $A \xrightarrow{h} B \xrightarrow{k} C$ we have that

$$\pi_0(k \circ h) = \pi_0(k) \circ \pi_0(h).$$

(iii) $\pi_0(\mathrm{id}_X) = \mathrm{id}_{\pi_0(X)}$.

Proof. Choose $x, x' \in X$ such that $[x] = [x']$. Then there exists a path γ from x to x' in X. Now $f \circ \gamma$ is a path from $f(x)$ to $f(x')$ hence they are in the same path-component so $\pi_0(f)$ is well-defined.

For (i), let $H : X \times I \to Y$ be a homotopy from f to g. Choose any $x \in X$. The map $H(x, \cdot)$ is a path from $f(x)$ to $g(x)$ so $[f(x)] = [g(x)]$ and hence $\pi_0(f) = \pi_0(g)$. (ii) and (iii) are obvious. □

Corollary. If $f : X \to Y$ is a homotopy equivalence then $\pi_0(f)$ is bijective.

Proof. Let $g : Y \to X$ be the homotopy inverse of f so $fg \simeq \mathrm{id}_Y$ hence $\pi_0(f)\pi_0(g) = \pi_0(\mathrm{id}_Y) = \mathrm{id}_{\pi_0(Y)}$. Similary $gf \simeq \mathrm{id}_X$ gives the other direction. Hence $\pi_0(f)$ is a bijection. □

Example. Consider the space $\{0, 1\}$ equipped with the discrete topology. Let's prove it is not contractible. If it were, then we'd have $\pi_0(\{0,1\}) \simeq \pi_0(*)$ (the one point space). This is false because $\{0, 1\}$ has two path-components.

Example. $\mathbb{R}^2$ does not retract to $A = \{(-1,0),(1,0)\}$. Suppose that $r : \mathbb{R}^2 \to A$ is a retraction. Then the composition

$$\pi_0(A) \overset{\pi_0(\tau)}{\to} \pi_0(\mathbb{R}^2) \overset{\pi_0(r)}{\to} \pi_0(A)$$

has to be the identity. This is impossible.

Definition. (Homotopic paths) The paths $\gamma, \gamma' : I \to X$ are *homotopic as paths* if they are homotopic rel $\{0,1\} \subseteq I$.

Lemma. If $\gamma_0, \gamma_1 : x_0 \rightsquigarrow x_1$ and $\gamma_0', \gamma_1' : x_1 \rightsquigarrow x_2$ are homotopic then $\gamma_0 \circ \gamma_0' \simeq \gamma_1 \circ \gamma_1' : x_0 \rightsquigarrow x_2$.

Proof. Let H be the homotopy of paths from γ_0 to γ_1 and let H' be the homotopy of paths from γ_0' to γ_1'. Define

$$H''(s,t) = \begin{cases} H(2s,t) & 0 \le s \le \frac{1}{2} \\ H'(2s-1,t) & \frac{1}{2} \le s \le 1 \end{cases}$$

which is continuous by the gluing lemma. At $H''(s,0) = (\gamma_0 \circ \gamma_0')(s)$ and $H''(s,1) = \gamma_1 \circ \gamma_1'$. So H'' is a homotopy from $\gamma_0 \circ \gamma_0'$ to $\gamma_1 \circ \gamma_1'$.

Proposition. Let $\gamma_0 : x_0 \rightsquigarrow x_1, \gamma_1 : x_1 \rightsquigarrow x_2, \gamma_2 : x_2 \rightsquigarrow x_3$ be paths in X. Then
(i) $(\gamma_0 \circ \gamma_1) \circ \gamma_2 \simeq \gamma_0 \circ (\gamma_1 \circ \gamma_2)$;
(ii) $\gamma_0 \circ c_{x_1} \simeq \gamma_0 \simeq c_{x_0} \circ \gamma_0$;
(iii) $\gamma_0 \gamma_0^{-1} \simeq c_{x_0}$, $\gamma_0^{-1} \circ \gamma_0 \simeq c_{x_1}$.

Proof. From these, we arrive at the following formulae.

(i) The homotopy

$$H(s,t) = \begin{cases} \gamma_0\left(\dfrac{4s}{t+1}\right) & 0 \le s \le \dfrac{t+1}{4}, \\ \gamma_1(4s-1-t) & \dfrac{t+1}{4} \le s \le \dfrac{t+2}{4}, \\ \gamma_2\left(1 - \dfrac{4(1-s)}{2-t}\right) & \dfrac{t+2}{4} \le s \le 1. \end{cases}$$

goes from $(\gamma_0 \cdot \gamma_1) \cdot \gamma_2$ to $\gamma_0 \cdot (\gamma_1 \cdot \gamma_2)$ as paths from x_0 to x_3.

(ii) The homotopy

$$H(s,t) = \begin{cases} \gamma_0\left(\dfrac{2s}{t+1}\right) & 0 \le s \le \dfrac{t+1}{2}, \\ x_1 & \dfrac{t+1}{2} \le s \le 1, \end{cases}$$

goes from $\gamma_0 \cdot c_{x_1}$ to γ_0 as paths from x_0 to x_1.

(iii) The homotopy

$$H(s,t) = \begin{cases} \gamma_0(2s) & 0 \le s \le \dfrac{1-t}{2}, \\ \gamma_0(1-t) & \dfrac{1-t}{2} \le s \le \dfrac{1+t}{2}, \\ \gamma_0(2-2s) & \dfrac{1+t}{2} \le s \le 1. \end{cases}$$

goes from $\gamma_0 \cdot \overline{\gamma_0}$ to c_{x_0} as paths from x_0 to x_0.

2.3 The Fundamental Group

Proposition. Let X be a space, $x_0 \in X$. Define

$$\pi_1(X, x_0) = \{\text{loops in } X \text{ based at } x_0\}/\text{homotopy}.$$

Then the rule $[\gamma] \circ [\gamma'] = [\gamma \circ \gamma']$ defines a group structure on $\pi_1(X, x_0)$. The identity is paths homotopic to the constant path.

Proof. The previous proposition.

Definition. (Fundamental Group) We call $\pi_1(X, x_0)$ the *fundamental group* of X at x_0.

Definition. (Based space) A *based space* is a pair (X, x_0) where x_0 is called the *basepoint*. Given based spaces $(X, x_0), (Y, y_0)$ a *based map* between them is a map $f : X \to Y$ such that $f(x_0) = y_0$. A *based homotopy* is a homotopy relative to the basepoint.

Proposition. For a based map $f : (X, x_0) \to (Y, y_0)$ there is a well-defined associated group homomorphism

$$\pi_1(f) : \pi_1(X, x_0) \to \pi_1(Y, y_0)$$

given by $\pi_1(f)[\gamma] \to [f \circ \gamma]$, with the following

(i) If f is based homotopic to g then $\pi_1(f) = \pi_1(g)$.
(ii) $\pi_1(f \circ g) = \pi_1(f) \circ \pi_1(g)$.
(iii) $\pi_1(\mathrm{id}_X) = \mathrm{id}_{\pi_1(X,x_0)}$.

Proof. $\pi_1(f)$ is well-defined because if $\gamma \simeq \gamma'$ then $f \circ \gamma \simeq f \circ \gamma'$.

We know that $f \circ c_{x_0} = c_{f(x_0)} = c_{y_0}$ so $\pi_1(f)$ preserves the identity element of the group. We also have that

$$f \circ (\gamma \cdot \gamma') = (f \circ \gamma) \cdot (f \circ \gamma')$$

so $\pi_1(f)$ is multiplicative, hence $\pi_1(f)$ is a group homomorphism.

For (i) if f, g are homotopic rel x_0 then $f \circ \gamma \simeq g \circ \gamma$ so $\pi_1(f) = \pi_1(g)$. For (ii) since $\pi_(f \circ g)[\gamma] = [f \circ g \circ \gamma] = \pi_(f)[g \circ \gamma] = \pi_1(f) \circ \pi_1(g)$. Part (iii) is obvious. □

For ease of notation, we'll just write f_* to mean $\pi_1(f)$.

Proposition. Let $u : x_0 \rightsquigarrow x_1$ be a path in X. Then it induces a group homomorphism

$$\begin{aligned} u_\# : \pi_1(X, x_0) &\to \pi_1(X, x_1) \\ [\gamma] &\to [u^{-1} \cdot \gamma \cdot u] \end{aligned}$$

satisfying

(i) $u \simeq u' \implies u_\# = u'_\#$;
(ii) $(c_{x_0})_\# = \mathrm{id}_{\pi_1(X,x_0)}$;
(iii) $v : x_1 \rightsquigarrow x_2 \implies (u \cdot v)_\# = v_\# u_\#$;
(iv) If $x_0 = x_1$, then $u_\#$ is conjugation by $[u]$;
(v) $f : X \to Y$, then we have

$$(f \circ u)_\# \circ f_* = f_* \circ u_\#.$$

Proof. Parts (i)-(iv) are obvious. We'll just prove (v).

We have that

$$\begin{aligned} ((f \circ u)_\# \circ f_*)[\gamma] &= (f \circ u)_\#[f \circ \gamma] \\ &= [(f \circ u)^{-1} \cdot (f \circ \gamma) \cdot (f \circ u)] \\ &= [f \circ (u^{-1} \cdot \gamma \cdot u)] \\ &= f_*[u^{-1} \cdot \gamma \cdot u] \\ &= f_* u_\#[\gamma]. \quad \square \end{aligned}$$

Suppose $f, g : X \to Y$ with $x_0 \in X$ and $H : X \times I \to Y$ is a homotopy from f to g. Then $H(x_0, \cdot) : I \to Y$ is a path from $f(x_0)$ to $g(x_0)$. Write this path as u. This sets us up to prove the following lemma

Lemma. For u, f, g given as above,

$$u_\# \circ f_* = g_*.$$

Proof. Choose any loop γ at x_0. Consider the composition

$$F : I \times I \xrightarrow{\gamma \times \mathrm{id}_I} X \times I \xrightarrow{H} Y$$

so $F(s,t) = H(\gamma(s), t))$. $\square$

Theorem. Let $f : X \to Y$ and choose $x_0 \in X$. Then $f_* : \pi_1(X, x_0) \to \pi_1(Y, f(x_0))$. If f is a homotopy equivalence, then f_* is a group isomorphism.

Proof. Let $g : Y \to X$ be a homotopy inverse for f such that $f \circ g \simeq \mathrm{id}$ and $g \circ f \simeq \mathrm{id}$. Choose homotopies H from fg to id and H' from gf to id. Let

$$u'(t) = H'(x_0, 1 - t) : I \to X.$$

Then by the previous lemma with the identity map we get that $u'_\# = (g \circ f)_* = g_* \circ f_*$. We know that $u'_\#$ is an isomorphism hence $g_* \circ f_*$ and $f_* \circ g_*$ are both isomorphisms (by doing the

same thing the otherway around). Hence since the first result gives that f_* is injective and g_* is surjective, and the second result gives that the otherway around, we get that both functions are isomorphisms, namely that f_* is an isomorphism. □

Definition. (Simply connected) A space X is *simply connected* if it is path-connected and satifies $\pi_1(X, x_0) = 1$ for all $x_0 \in X$.

Remark. From the previous proposition, since X is path-connected, we can connected every two points with a path, so $\pi_1(X, x_0) \cong \pi_1(X, x_1)$, hence it's equivalent that $\pi_1(X, x_0) = 1$ for a $x_0 \in X$ is enough for equivalence of simply connectedness.

Example. Any contractible space is simply connected. This is clear since if X is contractible then $X \simeq \{*\}$ so $\pi_1(X, x_0) \cong \pi_1(*, *) = 1$.

Lemma. A space X is simply connected if and only if for all x_0, x_1 there is a unique homotopy class of paths from x_0 to x_1.

Proof. The forward direction is obvious just from taking $x_0 = x_1$ where the result follows directly.

Instead if we suppose that X is simply connected and $\pi_1(X, x_1) = 1$. Since X is path-connected, there is a path from x_0 to x_1. Choose two paths γ and γ' from x_1 to x_1. Then $[\gamma^{-1} \cdot \gamma'] \in \pi_1(X, x_1)$. So $\gamma^{-1} \cdot \gamma' \simeq c_{x_1}$. Hence there is a homotopy from $\gamma \cdot \gamma^{-1} \cdot \gamma'$ to $\gamma \cdot c_{x_1}$. Since we can reduce the path $\gamma \cdot \gamma^{-1}$ to a point, we have that $\gamma' \simeq \gamma$. □

3 Covering Space

Definition. (Covering map) A map $f : \tilde{X} \to X$ is a *covering map* if it is surjective and for all $p \in X$ there is an open neighbourhood $U_p \subset X$ such that $f^{-1}(U_p) \simeq \bigsqcup_{\alpha \in I} V_\alpha$ where each

$$f|_{V_\alpha} : V_\alpha \to U_p$$

is a homeomorphism. We say that $\tilde{X}$ is the covering space of X.

Example. Clearly homeomorphism are covering maps

Example. If $p : \tilde{X} \to X$ and $q : \tilde{Y} \to Y$ are covering maps then $p \times q : \tilde{X} \times \tilde{Y} \to X \times Y$ is a covering map.

Example. Consider $S^1 = \{z \in \mathbb{C} : |z| = 1\} \subseteq \mathbb{C}$. Take $p(t) = e^{2\pi i t} : \mathbb{R} \to S^1$.

To prove that p is a covering map, consider the set $U_{y>} = \{x + iy \in S^1 : y > 0\}$, the open upper semicircle. Then

$$p^{-1}(U_{y>0}) = \bigsqcup_{j \in \mathbb{Z}} \left(j, j + \frac{1}{2}\right) \subset \mathbb{R}$$

which is the disjoint union of $\mathbb{Z}$ many open intervals. Furthermore

$$p|_{(j, j+\frac{1}{2})} : \left(j, j + \frac{1}{2}\right) \to U_{y>0}$$

is a continuous bijection with inverse

$$U_{y>0} \longrightarrow \left(j, j+\frac{1}{2}\right)$$
$$x+iy \longrightarrow j+\cos^{-1}(x)/2\pi.$$

Similarly with $U_{y<0}, U_{x>0}, U_{x<0}$ these open sets cover S^1. Thus p is a covering map.

Example. Consider $p(z) = z^n : S^1 \to S^1$ and denote $\eta = e^{\frac{2\pi i}{n}}$ and let ζ be some nth root of y. Then we have that

$$p^{-1}(y) = \{\zeta, \eta\zeta, \eta^2\zeta, \ldots, \eta^{n-1}\zeta\}$$

and $p^{-1}(S^1 \setminus \{y\})$ is the complement of this set. Now we can define

$$V_0 = \{z \in S^1 \mid 0 < \arg(z/\eta) < 2\pi/n\}$$

and $V_i = V_0 \cdot \eta^i$ for $0 \leq i < n$, so we get that

$$p^{-1}(S^1 \setminus \{y\}) = \bigsqcup_{i=0}^{n-1} V_i.$$

Then $p\,|_{V_i} \colon V_i \to S^1 \setminus \{y\}$ is a homeomorphism and

$$S^1 = \bigcup_{y \in S^1} S^1 \setminus \{y\}$$

therefore p is a covering map.

Example. Consider $S^2 = \{v \in \mathbb{R}^3 \mid ||v|| = 1\} \subseteq \mathbb{R}^3$. We can define an equivalence relation $x \sim -x$ on S^2 which defines a quotient space $\mathbb{RP}^2 = S^2/\sim$. Consider the quotient map

$$p : S^2 \longrightarrow \mathbb{RP}^2$$
$$x \longrightarrow [x]$$

which we claim is a covering map. Let $V = \{(x, y, z|) \in S^2 \mid z \neq 0\}$ then $V \simeq V_{z>0} \sqcup V_{z<0}$ so we can define p on the restriction to each of these subspaces. We claim that

$$p\,|_{V_{z>0}} \colon V_{z>0} \to p(V)$$
$$p\,|_{V_{z<0}} \colon V_{z<0} \to p(V)$$

are homeomorphisms. Similarly define $V_x, V_y \subset S^2$ then $p(V_x), p(V_y), p(V_z)$ are evenly covered and forms san open cover of $\mathbb{RP}^2$. Hence p is a covering map.

Definition. (Lifting) Let : $p : \tilde{X} \to X$ be a covering map and $f : Y \to X$ be any map. Then a map $\tilde{f} : Y \to \tilde{X}$ is a *lift* of f along p if $f = p \circ \tilde{f}$.

Lemma. (Uniqueness of lifts lemma) Let $p : \tilde{X} \to X$ be a covering map and $f : Y \to X$ be any map. Let $\tilde{f}_0, \tilde{f}_1$ be two lifts of f along p. Then the set $\{y \in Y : \tilde{f}_0(y) = \tilde{f}_1(y)\}$ is clopen in Y. Consequently if Y is connected then $S = Y$ or $S = \emptyset$.

Proof. First we'll show that S is open. Choose $y \in S$ and let U be an open neighbourhood of $f(s) \in X$ which is evenly covered, so $p^{-1}(U) = \bigsqcup_{\alpha \in I} V_\alpha$. Then $\tilde{f}_0(y)$ and $\tilde{f}_1(y)$ are equal so lie in the same V_β. Thus we have that on $N = \tilde{f}_0^{-1}(V_\beta) \cap \tilde{f}_1^{-1}(V_\beta)$ we have that

$$p|_{V_\beta} \circ \tilde{f}_0 \mid_N = f \mid_N = p \mid_{V_\beta} \circ \tilde{f}_1 \mid_N$$

but since $p \mid V_\beta$ is a homeomorphism, it has an inverse so $\tilde{f}_0 \mid_N = \tilde{f}_1 \mid_N$ and $y \in N \subset S$, so S is open.

Now we have to show that S is closed. We'll show that $Y \setminus S$ is open. Suppose that $y \notin S$, so $\tilde{f}_0(y) \neq \tilde{f}_1(y)$. Choose an evenly covered open neighbourhood U at $f(y)$ and identify $p^{-1}(U) = \bigsqcup_{\alpha \in I} V_\alpha$ so $\tilde{f}_0(y) \in V_\alpha$ and $\tilde{f}_1(y) \in V_\beta$ for some $\alpha, \beta \in I$ since V_α is mapped homeomorphically to U and the covering maps differ, we must have that $\alpha \neq \beta$. So $N = \tilde{f}_1^{-1}(V_\beta) \cap \tilde{f}_0^{-1}(V_\alpha)$ is an open neighbourhood of y so must intersect S by the definition of closure. But then V_α and V_β must intersect which is a contradiction.

Lemma. (Homotopy lifting lemma) Let $p : \tilde{X} \to X$ be a covering map, let Y be an arbitrary topological space, and let $H : Y \times I \to X$ be a homotopy from f_0 to f_1, i.e. $H(-,0) = f_0$ and $H(-,1) = f_1$. Let $\tilde{f}_0 : Y \to \tilde{X}$ be a lift of f_0. Then there exists a unique homotopy $\tilde{H} : Y \times I \to \tilde{X}$ such that
(i) $\tilde{H}(-,0) = \tilde{f}_0$;
(ii) $\tilde{H}$ is a lift of H along p, i.e. $p \circ \tilde{H} = H$.
In particular, $\tilde{H}$ is a homotopy from $\tilde{f}_0$ to the lift $\tilde{f}_1 := \tilde{H}(-,1)$ of f_1.

Proof. Throughout, $I = [0,1]$. We use the *uniqueness of lifts* lemma in the following form: if Z is a connected space and $g_1, g_2 : Z \to \tilde{X}$ are two lifts along p of the same map $Z \to X$ which agree at one point of Z, then $g_1 = g_2$. We only ever apply it with $Z = I$; in particular no connectedness (or local path connectedness) of Y is needed anywhere below.

Step 1: uniqueness. Let $\tilde{H}_1$ and $\tilde{H}_2$ both satisfy (1) and (2). Fix $y \in Y$. The paths $t \mapsto \tilde{H}_1(y,t)$ and $t \mapsto \tilde{H}_2(y,t)$ in $\tilde{X}$ are both lifts of the path $t \mapsto H(y,t)$ in X, and by (1) both start at $\tilde{f}_0(y)$. Since I is connected, uniqueness of lifts gives $\tilde{H}_1(y,t) = \tilde{H}_2(y,t)$ for all $t \in I$. As $y \in Y$ was arbitrary, $\tilde{H}_1 = \tilde{H}_2$.

This argument works for every space Y, so it applies in particular to any open subset $Y' \subseteq Y$ with the restricted data $H|_{Y' \times I}$ and $\tilde{f}_0|_{Y'}$.

Step 2: local existence. We claim that for every $y_0 \in Y$ there exist an open neighbourhood U_{y_0} of y_0 in Y and a continuous map $\tilde{H}_{y_0} : U_{y_0} \times I \to \tilde{X}$ such that $p \circ \tilde{H}_{y_0} = H|_{U_{y_0} \times I}$ and $\tilde{H}_{y_0}(-,0) = \tilde{f}_0|_{U_{y_0}}$.

(2a) Choosing a partition. Fix $y_0 \in Y$. For each $t \in I$ the point $H(y_0,t)$ has an evenly covered open neighbourhood in X. By continuity of H there are an open neighbourhood W_t of y_0 in Y and an open neighbourhood N_t of t in I such that $H(W_t \times N_t)$ is contained in an evenly covered open subset of X. The sets $\{N_t\}_{t \in I}$ form an open cover of the compact space I, so there is a finite subcover $N_{s_1}, \ldots, N_{s_k}$. Put

$$W = W_{s_1} \cap \cdots \cap W_{s_k},$$

an open neighbourhood of y_0. By the Lebesgue number lemma there is an $N \geq 1$ such that each interval $[\frac{i}{N}, \frac{i+1}{N}]$ lies in some N_{s_j}. Write $t_i = \frac{i}{N}$. Then for each $i = 0, \ldots, N-1$ there is an evenly

covered open subset $U_i \subseteq X$ with

$$H\left(W \times [t_i, t_{i+1}]\right) \subseteq H(W_{s_j} \times N_{s_j}) \subseteq U_i.$$

Write

$$p^{-1}(U_i) = \bigsqcup_{\alpha \in A_i} V_{i,\alpha}, \qquad p\,|_{V_{i,\alpha}} \colon V_{i,\alpha} \xrightarrow{\sim} U_i$$

where the $V_{i,\alpha}$ are pairwise disjoint open subsets of $\tilde{X}$ and $\xrightarrow{\sim}$ means a homeomorphism.

(2b) Inductive construction with shrinking. We show by induction on $i = 0, \dots, N$ that there exist an open neighbourhood W_i' of y_0 with $W_i' \subseteq W$ and a continuous map $G_i : W_i' \times [0, t_i] \to \tilde{X}$ such that $p \circ G_i = H|_{W_i' \times [0,t_i]}$ and $G_i(y, 0) = \tilde{f}_0(y)$ for all $y \in W_i'$.

For $i = 0$ take $W_0' = W$ and $G_0(y, 0) = \tilde{f}_0(y)$; then $p \circ G_0 = f_0|_W = H(-, 0)|_W$.

Suppose W_i' and G_i are constructed for some $i < N$. The point $G_i(y_0, t_i)$ satisfies $p(G_i(y_0, t_i)) = H(y_0, t_i) \in U_i$, so it lies in $p^{-1}(U_i)$, hence in exactly one sheet $V = V_{i,\alpha}$. Since $y \mapsto G_i(y, t_i)$ is continuous and V is open,

$$W_{i+1}' = \{y \in W_i' \mid G_i(y, t_i) \in V\}$$

is an open neighbourhood of y_0. Define $G_{i+1} : W_{i+1}' \times [0, t_{i+1}] \to \tilde{X}$ by

$$G_{i+1}(y, t) = \begin{cases} G_i(y, t) & t \in [0, t_i], \\ (p\,|_V)^{-1}\left(H(y, t)\right) & t \in [t_i, t_{i+1}]. \end{cases}$$

The second formula makes sense because $H(y, t) \in U_i$ for $y \in W \supseteq W_{i+1}'$ and $t \in [t_i, t_{i+1}]$. The two formulas agree at $t = t_i$: indeed $G_i(y, t_i) \in V$ and $p(G_i(y, t_i)) = H(y, t_i)$, so $(p\,|_V)^{-1}(H(y, t_i)) = G_i(y, t_i)$. Both pieces are continuous on closed subsets $W_{i+1}' \times [0, t_i]$ and $W_{i+1}' \times [t_i, t_{i+1}]$ whose union is the whole domain, so G_{i+1} is continuous by the pasting lemma. Clearly $p \circ G_{i+1} = H$ and $G_{i+1}(-, 0) = \tilde{f}_0$ on W_{i+1}', which completes the induction.

Finally set $U_{y_0} = W_N'$ and $\tilde{H}_{y_0} = G_N$. Note that we shrank the neighbourhood only finitely many times, so U_{y_0} is still an open neighbourhood of y_0; this is exactly where one would otherwise want Y to be (locally) connected, and the shrinking trick avoids it.

Step 3: the local lifts are compatible. Let $y_0, y_1 \in Y$ and $Y' = U_{y_0} \cap U_{y_1}$, which is open. If $Y' = \emptyset$ there is nothing to show. Otherwise $\tilde{H}_{y_0}|_{Y' \times I}$ and $\tilde{H}_{y_1}|_{Y' \times I}$ are both lifts of $H|_{Y' \times I}$ whose restriction to $Y' \times \{0\}$ is $\tilde{f}_0|_{Y'}$. By the uniqueness in Step 1 (applied to Y'), $\tilde{H}_{y_0} = \tilde{H}_{y_1}$ on $(U_{y_0} \cap U_{y_1}) \times I$.

Step 4: gluing. Define $\tilde{H} : Y \times I \to \tilde{X}$ by

$$\tilde{H}(y, t) = \tilde{H}_{y_0}(y, t) \quad \text{for any } y_0 \in Y \text{ with } y \in U_{y_0}.$$

(Such a y_0 exists, e.g. $y_0 = y$.) This is well defined by Step 3. The sets $U_{y_0} \times I$ form an open cover of $Y \times I$ and $\tilde{H}$ restricts to the continuous map $\tilde{H}_{y_0}$ on each of them, so $\tilde{H}$ is continuous, since continuity is a local property. Moreover $p \circ \tilde{H} = H$ and $\tilde{H}(-, 0) = \tilde{f}_0$ hold on each $U_{y_0} \times I$, hence on $Y \times I$. This proves existence, and uniqueness was shown in Step 1. □

Corollary. (Path lifting) For a path $\gamma : I \to X$ choose $\tilde{x}_0 \in \tilde{X}$ such that $p(\tilde{x}_0) = \gamma(0)$. Then there exists a unique path $\tilde{\gamma} : I \to \tilde{X}$ such that $\tilde{\gamma}(0) = \tilde{x}_0$ and $p \circ \tilde{\gamma} = \gamma$.

Proof. Take $Y = \{*\}$ and apply the homotopy lifting lemma.

Corollary. Let $\gamma, \gamma' : I \to X$ be paths in X from x_0 to x_1 and $\tilde{\gamma}, \tilde{\gamma}' : I \to \tilde{X}$ be lifts starting at $\tilde{x}_0 \in p^{-1}(x_0)$. If $\gamma \simeq \gamma'$ as paths, then $\tilde{\gamma} \simeq \tilde{\gamma}'$ as paths, in particular $\tilde{\gamma}(1) \simeq \tilde{\gamma}'(1)$.

Proof. Let $H : I \times I \to X$ be a homotopy of paths from γ to γ' and lifts strating at $\tilde{\gamma}$ to obtain a homotopy $\tilde{H} : I \times I \to \tilde{X}$ such that $\tilde{H}(\cdot, 0) = c_{\tilde{x}_0}$. We have to show that $\tilde{H}(\cdot, 1)$ is a constant path.

Observe that $\tilde{H}(\cdot, 1)$ is a path in $\tilde{X}$, $p \circ \tilde{H}(\cdot, 1) = H(\cdot, 1)$ is constant. By uniqueness, $\tilde{H}(\cdot, 1)$ is constant. □

Suppose we have $p : \tilde{X} \to X$ being a covering map and a point $x_0 \in X$ with $\tilde{x}_0 \in p^{-1}(x_0)$. For a path $u : x_0 \rightsquigarrow x_1$ in X, the first corollary tells us that there exists a lift $\tilde{u}$ of u with domain $\tilde{x}_0$. Then $\tilde{x}_0 \to \tilde{u}(1)$ is a well-defined function we denote

$$u_* : p^{-1}(x_0) \to p^{-1}(x_1).$$

By the second corollary u_* depends only on the homotopy class at u.

Corollary. We have the following:
(i) If $u : x_0 \rightsquigarrow x_1$ and $u' : x_1 \rightsquigarrow x_2$ then $(u \cdot u')_* = u'_* \circ u_*$;
(ii) u_* is a bijection;
(iii) If X is path connected then $|p^{-1}(x_0)| = |p^{-1}(x_1)|$.

Proof. We can see that (ii) and (iii) following from (i), so we'll just prove (i). Choose a lift $\tilde{u} : \tilde{x}_0 \to \tilde{x}_1$ of u and $\tilde{u}' : \tilde{x}_1 \to \tilde{x}_2$ of u'. Then $\tilde{u} \cdot \tilde{u}' : \tilde{x}_0 \to \tilde{x}_2$ is a lift of $u \cdot u'$. Thus $(u \cdot u')_*(\tilde{x}_0) = \tilde{x}_2 = u'_*(\tilde{x}_1) = u'_*(u_*(\tilde{x}_0)) = (u'_* \circ u_*)(\tilde{x}_0)$. □

Lemma. Let $p : \tilde{X} \to X$ be a covering map. Choose $x_0 \in X$, $\tilde{x}_0 \in p^{-1}(x_0)$. Then

$$p_* : \pi_1(\tilde{X}, \tilde{x}_0) \to \pi_1(X, x_0)$$

is injective.

Proof. Choose a loop γ in $\tilde{X}$, based at $\tilde{x}_0$ such that $[\gamma] \in \ker p_*$. Hence $[p \circ \gamma] = 0$. So there is a homotopy $H : I \times I \to X$ from $p \circ \gamma$ to c_{x_0}. By the homotopy lifting lemma, H admits a lift $\tilde{H}$ which is a homotopy γ to some lift of c_{x_0}. By uniqueness of lifts, the lift of c_{x_0} is $c_{\tilde{x}_0}$ so $[\gamma] = [c_{\tilde{x}_0}] = 0 \in \pi_1(\tilde{X}, \tilde{x}_0)$ so p_* is injective.

Definition. (n-sheeted cover) For a covering map $p : \tilde{X} \to X$ we say that p is an *n-sheeted cover* for $n \in \mathbb{N} \cup \{\infty\}$ if each $p^{-1}(x)$ has cardinality n.

Definition. (Universal cover) For a covering map $p : \tilde{X} \to X$ we say that p is a *universal* if $\tilde{X}$ is *simply connected* and we say that $\tilde{X}$ is a *universal cover* of X.

From the above lemma we have an action of $\pi_1(X, x_0)$ acting on $p^{-1}(x_0)$ by the action for some

$[\gamma]$ and $\tilde{x}_0$ a point in the fiber, we have

$$\tilde{x}_0 \cdot [\gamma] = \tilde{\gamma}(1)$$

where $\tilde{\gamma}$ is the unique lift of γ. This is well-defined by the uniqueness of lifts for homotopically equivalent loops (up to homotopy).

Lemma. Let $p : \tilde{X} \to X$ be a covering map and X path-connected. Then

(i) $\tilde{X}$ is path-connected if and only if the action of $\pi_1(X, x_0)$ on $p^{-1}(x_0)$ is transitive.

(ii) The stabiliser for a point $y_0 \in p^{-1}(x_0)$ in the fibre is the subgroup

$$\mathrm{im}(p_* : \pi_1(\tilde{X}, y_0) \to \pi_1(X, x_0)) \leq \pi_1(X, x_0)$$

(iii) If the covering map is universal then the action is simply transitive.

(iv) If $\tilde{X}$ is path-connected then there is a bijection

$$\frac{\pi_1(X, x_0)}{p_*\pi_1(\tilde{X}, y_0)} \longrightarrow p^{-1}(x_0)$$

induced by action on the point y_0.

Proof.

(i) Take $y_0, y_1 \in p^{-1}(x_0)$ elements of the fibre. Then by assumption they are path-connected by some path $\gamma : I \to \tilde{X}$. So $p \circ \gamma : I \to X$ is a loop in X which is based at x_0 and lifts to γ, starting at y_0. The action of $[p \circ \gamma] \in \pi_1(X, x_0)$ takes y_0 to y_1. Since y_0 and y_1 were arbitrary action is transitive.

Conversely suppose that $y_0, z_0 \in \tilde{X}$ lie in different path-components. Since X is path connected, choose paths γ_y and γ_z from $p(y_0)$ to x_0 and from $p(z_0)$ to x_0 and lift them into $\tilde{X}$. These paths therefore end at $y_1, z_1 \in p^{-1}(x_0)$ which must again be in different path components. But $\pi_1(X, x_0)$ acts transitively on $p^{-1}(x_0)$ so there is a $[\gamma'] \in \pi_1(X, x_0)$ such that $z_1 \bullet [\gamma'] = y_1$, i.e the lift of γ' starting at z_1 ends at y_1, so they're path-connected so y_0 and z_0 are path-connected which is a contradiction.

(ii) Suppose that $y_0 \bullet [\gamma] = y_0$ so the lift $\tilde{\gamma}$, of γ starting at y_0 also ends at y_0. So $\tilde{\gamma}$ is a loop based at y_0. We have that $\gamma = p \circ \tilde{\gamma}$ i.e that $p_*([\tilde{\gamma}]) = [\gamma] \in \mathrm{im}(p_*)$.

Conversely suppose that $[\gamma] \in \mathrm{im}(p_*)$. So there exists a $[\tilde{\gamma}] \in \pi_1(\tilde{X}, y_0)$ such that $\gamma = p \circ \tilde{\gamma}$ starting at y_0. It ends at y_0 too so we have that $y_0 \bullet [p \circ \gamma] = y_0$ hence it is in the stabiliser of y_0.

(iii) If $\tilde{X}$ is a universal cover then it is simply connected so $\pi_1(\tilde{X}, y_0) \cong \{*\}$. So the image of p_* is trivial, hence by (ii) the stabiliser for a point y_0 in the fibre is trivial, so all stabiliser are trivial and the action is free. Since it is free and transitive by (i) it is simply transitive (regular).

(iv) Apply orbit-stabiliser. □

Remark. In this case it's good to think of "simply transitive" as meaning for every pair of points in the fibre, there is *exactly* one element taking one point to the other.

3.1 The Fundamental Group of the Circle

Theorem. We have that
$$\pi_1(S^1, 1) \cong \mathbb{Z}.$$

Proof. Consider the covering map
$$p(t) = e^{2\pi it} : \mathbb{R} \to S^1$$
which we proved was a covering map. We know that $\mathbb{R}$ is contractible and hence it is simply connected so p is universal (since S^1 and $\mathbb{R}$ is path-connected). Thus (taking arbitrary base points) we have that $\pi_1(S^1, 1)$ acts on $p^{-1}(1) = \mathbb{Z}$ and is simply transitive.

Consider the loop $\gamma(t) = e^{2\pi it} : I \to S^1$ which is a loop based at $1 \in S^1$. So we have $[\gamma] \in \pi_1(S^1, 1)$. Consider the induced action $\langle [\gamma] \rangle$ on $p^{-1}(1)$. We claim that is this also simply transitive, in which case it will follow that $\langle [\gamma] \rangle = \pi_1(S^1, 1) \cong \mathbb{Z}$.

Consider $[\gamma] \bullet n = \tilde{\gamma}(1)$ where $\tilde{\gamma}$ is a lift of γ such that $\tilde{\gamma}(0) = n$. We'll guess that
$$\tilde{\gamma}(t) = n + t \implies p \circ \tilde{\gamma} = e^{2\pi i(n+t)} = e^{2\pi it} = \gamma(t)$$
so by uniqueness this our unique lift starting at n. So $\tilde{\gamma}(1) = n + 1$, hence $[\gamma] \bullet n = n + 1$. So this action is simply transitive and hence
$$\pi_1(S^1, 1) \cong \mathbb{Z}. \quad \square$$

Theorem. The disc D^2 does not retract to it boundary $\partial D^2 = S^1$.

Proof. Suppose we have a retraction $r : D^2 \to S^1$ and the inclusion map $\iota : S^1 \hookrightarrow D^2$. Then $r \circ \iota = \mathrm{id}_{S^1}$ and we have the factorisation
$$\mathbb{Z} \cong \pi_1(S^1, 1) \xrightarrow{\iota_*} \pi_1(D^2, 1) \xrightarrow{r^*} \pi_1(S^1, 1) \cong \mathbb{Z}$$
of the identity map. But D^2 is simply connected so it's fundamental group is trivial so this is a contradiction. $\square$

This gives us a nice corollary.

Theorem. (Brouwer's fixed point theorem). For a continuous map $f : D^2 \to D^2$ there is some $x \in D^2$ such that $f(x) = x$, i.e f has a fixed point.

Proof. Suppose that f has no fixed points. Define a function $r : D^2 \to S^1$ sending x to the first point on S^1 hit by the ray from $f(x)$ to x which makes sense since $f(x)$ and x are distinct points by assumption. Now r is continuous since f is continuous. But we have a continuous retraction of D^2 to S^1 but this is impossible from the previous theorem. $\square$

Theorem. (Fundamental Theorem of Algebra) Any polynomial $f \in \mathbb{C}[X]$ that is non-vanishing on $\mathbb{C}$ is constant.

Proof. Write that $f(x) = x^n + a_1x^{n-1} + \cdots + a_n$. Take $r > \max(1, \sum_{i=1}^n |a_1|)$. Then $|z|^n > |a_1x^{n-1} + \cdots + a_n|$ whenever $|z| = r$. Consider the polynomial

$$f_t(x) = x^n + t(a_1x^{n-1} + \cdots + a_n)$$

so that f_t is non-vanishing on $|x| = r$. Define

$$F(s,t) = \frac{f_t(re^{2\pi is})/f_t(r)}{|f_t(re^{2\pi is})/f_t(r)|}$$

which is well-defined so this is a homotopy $I \times I \to \mathbb{C}$. F is a homotopy from $[s \to e^{2\pi ins}]$ to the loop $\left[s \to \frac{f(re^{2\pi is}/f(r)}{|f(re^{2\pi is}/f(r)|}\right]$ which we will call $\gamma_r(s)$. Take $r \to 0$ which goes towards the constant loop c_1 hence since F is a homotopy, $[s \to e^{2\pi ins}]$ is the identity in $\pi_1(S^1, 1)$. But it also represents n. Hence $n = 0$ so f is constant. □

3.2 The Construction of Universal Covers

We will assume X to be *locally path-connected* (i.e all points have an open neighbourhood that is path-connected.

Definition. (Semilocally Simply Connected) A path-connected and locally path-connected space X is *semilocally simply connected* if for all $p \in X$ there exists an open neighbourhood $U_p \subseteq X$ of p such that the map

$$\pi_1(U_p, p) \to \pi_1(X, p)$$

is trivial i.e every loop in U_p can be contracted to a point.

Remark. This is different than U_p just being simply connected, since we're allowing the contraction to happen anywhere in X, so the loop may move around to contract itself somewhere else.

Theorem. If X is path-connected, locally path-connected, and semilocally simply connected then X admits a universal cover.

Proof. (Non-examinable) We'll just sketch a proof out now. Observe that

$$\mathcal{O} = \{U \subset X \text{ open} \mid U \text{ is path connected } \pi_1(U, p) \to \pi_1(X, p) \text{ trivial}\}$$

is an open cover at X. Write $\mathcal{O} = \{U_\alpha\}_{\alpha \in I}$. Choose $p \in X$. For each $\alpha \in I$, take $x_\alpha \in U_\alpha$. Choose a path γ_α from p to x_α on X. Then for all $x \in U_\alpha$ choose any path $\gamma^0_{x,\alpha}$ which is a path from x_α to x and concatenate $\gamma_{x,\alpha} = \gamma_\alpha\gamma^0_{x,\alpha}$ from p to x (in X). Then $[\gamma_{x,\alpha}]$ depends only on x, α, γ_α. Now for each $\alpha, \beta \in I$, for all $x \in U_\alpha \cap U_\beta$ define

$$\varphi_{\alpha,\beta} : U_\alpha \cap U_\beta \to \mathrm{Sym}(\pi_1(X, p))$$

by

$$\varphi_{\alpha,\beta}(x)(g) = g \bullet [\gamma_{x,\alpha}\gamma^{-1}_{x,\beta}].$$

This function is locally trivial and thus constant. For $\alpha, \beta, \gamma \in I$ and $x \in U_\alpha \cap U_\beta \cap U_\gamma$ we get that

$$\varphi_{\beta\gamma}\varphi_{\alpha\beta}(x) = \varphi_{\alpha\gamma}(x).$$

Form

$$\tilde{X} = \left(\bigsqcup_{\alpha \in I} (U_\alpha \times \pi_1(X, p)) \right) / \sim$$

where $(x \in U_\alpha, g) \sim (x \in U_\beta, \varphi_{\alpha\beta}(x)(g))$ whenever $\alpha, \beta \in I$, $x \in U_\alpha \cap U_\beta$. We have a map

$$p : \tilde{X} \to X$$

taking $(x, g) \to x$. This is a universal covering map (proof omitted). □

3.3 The Galois Correspondence

We know that $p_* : \pi_1(\tilde{X}, \tilde{x}_0) \to \pi_1(X, x_0)$ is injective and we obtain a subgroup $p_*(\pi_1(\tilde{X}, \tilde{x}_0))$. We want a correspondance

$$\left\{ \begin{array}{c} \text{based covering maps} \\ p : (\widetilde{X}, \tilde{x}_0) \to (X, x_0) \end{array} \right\} \longrightarrow \{\text{subgroups of } \pi_1(X, x_0)\}$$

and

$$\{\text{covering maps } p : \widetilde{X} \to X\} \longrightarrow \left\{ \begin{array}{c} \text{conjugacy classes of} \\ \text{subgroups of } \pi_1(X, x_0) \end{array} \right\}.$$

and we wish to turn these into bijections by dividing out by an appropriate equuivalence relation on the left hand side to turn both of these functions into bijections.

Proposition. Suppose that X is path-connected, locally path-connected and semilocalyl simply connected. Then for any subgroup $H \leq \pi_1(X, x_0)$ there is a based covering map $p : (\tilde{X}, \tilde{x}_0) \to (X, x_0)$ such that $p_*(\pi_1(\tilde{X}, \tilde{x}_0)) = H$.

Proof. From the assumptions on X we know that X has a universal cover $\overline{p} : \overline{X} \to X$. For each $\tilde{x} \in \overline{X}$ choose a path $\tilde{\gamma}$ from $\tilde{x}_0$ to $\tilde{x}$. Write $\gamma = p \circ \tilde{\gamma}$ and $x = \overline{p}(\tilde{x})$. Given any element $[\gamma_0] \in \pi_1(X, x_0)$, $\gamma_0\gamma$ is a path from x_0 to x. So there exists a lift $\widetilde{\gamma_0\gamma}$ from $\tilde{x}_0$ to some point. We define

$$[\gamma_0] \bullet \tilde{x} = \widetilde{\gamma_0\gamma}(1)$$

which defines a group action from the fundamental group on the universal covering space such that for all $g \in \pi(X, x_0)$ and $x \in \overline{X}$, $p(g\tilde{x}) = p(\tilde{x})$.

Choose any evenly covered (by $\overline{p}$) path-connected open subset $U \subset X$ then we claim that any element $g \in \pi_1(X, x_0)$ acts on $p^{-1}(U) \cong U \times F$, where F is the discrete space parameterising the fibre, by permuting F and fixing U i.e it is permuting the fibre sheets. Now we'll prove this claim.

Choose $x, y \in U$, $\alpha \in F$, $g \in \pi_1(X, x_0)$ and suppose that

$$g \bullet (x, \alpha) = (x, \beta)$$

with $\beta \in F$. Choose a path γ from x to y on U and take its lift, so we have the path $\tilde{\gamma}$ starting at (x, β). Then $\tilde{\gamma}$ is a path in the sheet $U \times \{\beta\}$ so write its endpoint as (y, β). So if we write $g = [\gamma_g]$ and choose paths γ_x taking x_0 to x and $\tilde{\gamma}_x$ taking x_0 to (x, α), then

$$\widetilde{\gamma_g\gamma_x}(0) = (x, \beta)$$

which gives that $g \cdot (y, \alpha) = \widetilde{\gamma_g \gamma_\alpha \gamma}(1) = \tilde{\gamma}(1) = (y, \beta)$. So $\pi_1(X, x_0)$ acting on $\overline{X}$ is continuous and $\overline{p} : \overline{X} \to X$ induces $p_H : \overline{X}/H \to X$.

Here $\pi_1(X, x_0)$ acts on $\overline{p}_H^{-1}(x_0)$ is right multiplication. The quotient by H is as the left. So the induced action of the fundamental group on the fibre $p_H^{-1}(x_0) = (H \setminus \pi_1(X, x_0))[\tilde{x}_0]$ so the stabiliser is H and hence this is the image of p_*. □

Proposition. (Based Uniqueness) Suppose that X is path-connected and locally path-connected. Suppose we have

$$p_1 : (\tilde{X}_1, \tilde{x}_2) \to (X, x_0) \qquad p_2 : (\tilde{X}_2, \tilde{x}_2) \to (X, x_0)$$

based covering maps. Then there exists a unique based homeomorphism

$$h : (\tilde{X}_1, \tilde{x}_1) \to (\tilde{X}_2, \tilde{x}_2)$$

such that $p_2 \circ h = p_1$ if and only if $\mathrm{im}(p_{1*}) = \mathrm{im}(p_{2*})$.

Proof. If h exists then h_* is an isomorphism so the result is obvious.

For the converse direction, choose any point $\tilde{x} \in \tilde{X}_1$ and path $\tilde{\gamma}$ from $\tilde{x}_1$ to $\tilde{x}$ on $\tilde{X}_1$. Consider $\gamma = p_1 \circ \tilde{\gamma}$ going from x_0 to $x_1 = p(\tilde{x})$. It has unique lift $\tilde{\gamma}'$ such that $\tilde{\gamma}(0) = \tilde{x}_2$ in $\tilde{X}_2$. Then we define $h(\tilde{x}) = \tilde{\gamma}'(1)$ and this is independent of the choice of $\tilde{\gamma}$. We also have to prove that h is continuous (it can be proved using path-connected, evenly covered open neighbourhoods. So we have constructed h uniquely. Similarly we have a h' which is our unique based homeomorphism from $(\tilde{X}_2, \tilde{x}_2)$ to $(\tilde{X}_1, \tilde{x}_1)$. So now $h' \circ h = \mathrm{id}_{\tilde{X}_1}$ by the uniqueness. Symmetrically we have that $h \circ h' = \mathrm{id}_{\tilde{X}_2}$ so h is a based homeomorphism. □

Remark. For this, we don't require X to be semilocally simply connected, i.e we don't need X to have a universal cover.